

Probability

1. A committee of three is to be selected from the nine members in a club.
 - a. How many different committees can be formed?
 - b. If there are five men in the club, what is the probability that the selected committee consists entirely of males?
2. The integers from 1 to 10 inclusive are written on ten separate pieces of paper. Four pieces of paper are drawn at random. Find the probability that:
 - a. The four numbers drawn are 1, 2, 3 and 6.
 - b. The number 9 is one of the numbers drawn
 - c. The number 8 is not drawn
 - d. The number 7 is drawn but the number 1 is not
3. A bag contains three red, seven yellow and five blue balls. If three balls are drawn from the bag simultaneously, find the probability that:
 - a. All three balls are yellow
 - b. All the balls are of the same colour,
 - c. There are two red balls and one blue ball
 - d. All the balls are of different colours
4. A sports committee of five members is to be chosen from eight AFL footballers and seven soccer players. Find the probability that the committee will contain:
 - a. Only AFL footballers
 - b. Only soccer players
 - c. Three soccer players and two AFL footballers
 - d. At least one soccer player
 - e. At most one soccer player
 - f. Ian, a particular soccer player
5. From a standard pack of 52 cards, three are selected at random. Find the probability that:
 - a. They are the jack of spades, the two of clubs and the seven of diamonds,
 - b. All three are aces
 - c. They are all diamonds
 - d. They are all of the same suit
 - e. They are all picture cards
 - f. Two are red and one is black
 - g. One is a seven, one is an eight and one is a nine
 - h. Two are 7s and one is a 6,
 - i. Exactly one is a diamond
 - j. At least two of them are diamonds.
6. Repeat the previous question if the cards are selected from the pack one at a time, and each card is replaced before the next one is drawn.
7. Three boys and three girls are to sit in a row. Find the probability that:
 - a. The boys and girls alternate
 - b. The boys and girls sit together
 - c. Two specific girls sit next to one another.
8. A family of five are seated in a row at the cinema. Find the probability that:
 - a. The parents sit on the end and the three children are in the middle
 - b. The parents sit next to one another
9. Six people, of whom Patrick and Jessica are two, arrange themselves in a row. Find the probability:
 - a. That Patrick and Jessica occupy the end positions
 - b. That Patrick and Jessica are not next to each other
10. The letters of PROMISE are arranged randomly in a row. Find the probability that:
 - a. The word starts with R and ends with S,
 - b. The letters P and R are next to one another
 - c. The letters P and R are separated by at least three letters,
 - d. The vowels and the consonants alternate
 - e. The vowels are together
11. The digits 3, 3, 4, 4, 4 and 5 are placed in a row to form a six-digit number. If one of these numbers is selected at random, find the probability that:
 - a. It is even
 - b. It ends in 5

- c. The 4s occur together
 - d. The number starts with 5 and then the 4s and 3s alternate
 - e. The 3s are separated by at least one other number
12. The letters of the word PRINTER are arranged in a row. Find the probability that:
- a. The word starts with the letter E
 - b. The letters I and P are next to one another
 - c. There are three letters between N and T
 - d. There are at least three letters between N and T.
13. The letters of KETTLE are arranged randomly in a row. Find the probability that:
- a. The two letters E are together
 - b. The two letters E are not together
 - c. The two letters E are together and the two letters T are together
 - d. The Es and Ts are together in one group
14. The letters of ENTERTAINMENT are arranged in a row. Find the probability that:
- a. The letters. E are together
 - b. Two Es are together and one is apart
 - c. All the letters E are apart
 - d. The word starts and ends with E.
15. A tank contains 20 tagged fish and 80 untagged fish. On each day, four fish are selected at random, and after noting whether they are tagged or untagged, they are returned to the tank. Answer the following questions, correct to three significant figures.
- a. What is the probability of selecting no tagged fish on a given day?
 - b. What is the probability of selecting at least one tagged fish on a given day?
 - c. Calculate the probability of selecting no tagged fish on every day for a week
 - d. What is the probability of selecting no tagged fish on exactly three of the seven days during the week?
16. A bag contains seven white and five black discs. Three discs are chosen from the bag. Find the probability that all three discs are black, if the discs are chosen:
- a. Without replacement
 - b. With replacement
 - c. So that after each draw the disc is replaced with one of the opposite colour
17. Six people are to be divided into two groups, each with at least one person. Find the probability
- a. That there will be three in each group
 - b. That there will be two in one group and four in the other
 - c. That there will be one group of five and an individual
18. A three-digit number is formed from the digits 3, 4, 5, 6 and 7 (no repetitions allowed). Find the probability that:
- a. The number is 473
 - b. The number is odd
 - c. The number is divisible by 5
 - d. The number is divisible by 3
 - e. The number starts with 4 and ends with 7
 - f. The number contains the digit 3
 - g. The number contains the digits 3 and 5
 - h. The number contains the digits 3 or 5
 - i. All digits in the number are odd
 - j. The number is greater than 500
19. The digits 1, 2, 3 and 4 are used to form numbers that may have 1, 2, 3 or 4 digits in them. If one of the numbers is selected at random, find the probability that:
- a. It has three digits
 - b. It is even
 - c. It is greater than 200
 - d. It is odd and greater than 3000
 - e. It is divisible by 3
20. a. A senate committee of five members is to be selected from six Labor and five Liberal senators. What is the probability that Labor a majority on the committee?
- b. The senate committee is to be selected from N Labor and five Liberal senators. Use trial and error to find the minimum value of N , given that the probability of Labor having a majority on the committee is greater than $\frac{3}{4}$.
21. Four basketball teams A, B, C and D each consist of ten players, and in each team, the players are numbered 1, 2, ... 9, 10. Five players are to be selected at random from the four teams. Find the probability that of the five players selected:

- a. Three are numbered 4 and two are numbered 9,
b. At least four are from the same team.
22. A poker hand of five cards is dealt from a standard pack of 52. Find the probability of obtaining:
- a. One pair ONLY
b. Two pairs
c. Three of a kind ONLY
d. Four of a kind
e. A full house (one pair and three of a kind)
f. A straight (five cards in sequence regardless of suit)
g. A flush, (five cards in sequence regardless of suit)
h. A royal flush (ten, jack, queen, king and ace in a single suit)
23. Four adults are standing in a room that has five exits. Each adult is equally likely to leave the room through any one of the five exits.
- a. What is the probability that all four adults leave the room via the same exit?
b. What is the probability that three particular adults use the same exit and the fourth adult uses a different exit?
c. What is the probability that any three of the four adults come out the same exit, and the remaining adult comes out a different exit?
d. What is the probability that no more than two adults come out the same exit?
24. a. Five diners in a restaurant choose randomly from a menu featuring five main courses. Find the probability that exactly one of the main courses is not chosen by any of the diners.
b. Repeat the question if there are n diners and a choice of n main courses.

Answer:

1. a. 84; b. $\frac{5}{42}$
2. a. $\frac{1}{210}$; b. $\frac{2}{5}$; c. $\frac{3}{5}$; d. $\frac{4}{15}$
3. a. $\frac{1}{13}$; b. $\frac{46}{455}$; c. $\frac{3}{91}$; d. $\frac{3}{13}$
4. a. $\frac{8}{429}$; b. $\frac{1}{143}$; c. $\frac{140}{429}$; d. $\frac{421}{429}$; e. $\frac{2}{11}$; f. $\frac{1}{3}$
5. a. $\frac{1}{22100}$; b. $\frac{1}{5525}$; c. $\frac{11}{850}$; d. $\frac{22}{425}$; e. $\frac{11}{1105}$; f. $\frac{13}{34}$; g. $\frac{16}{5525}$; h. $\frac{6}{5525}$; i. $\frac{741}{1700}$; j. $\frac{64}{425}$
6. a. $\frac{3}{70304}$; b. $\frac{1}{2197}$; c. $\frac{1}{64}$; d. $\frac{1}{16}$; e. $\frac{27}{2197}$; f. $\frac{3}{8}$; g. $\frac{6}{2197}$; h. $\frac{3}{2197}$; i. $\frac{27}{64}$; j. $\frac{5}{32}$
7. a. $\frac{1}{10}$; b. $\frac{1}{10}$; c. $\frac{1}{3}$
8. a. $\frac{1}{10}$; b. $\frac{2}{5}$
9. a. $\frac{1}{15}$; b. $\frac{2}{3}$
10. a. $\frac{1}{42}$; b. $\frac{2}{7}$; c. $\frac{2}{7}$; d. $\frac{1}{35}$; e. $\frac{1}{7}$
11. a. $\frac{1}{2}$; b. $\frac{1}{6}$; c. $\frac{1}{5}$; d. $\frac{1}{60}$; e. $\frac{2}{3}$
12. a. $\frac{1}{7}$; b. $\frac{2}{7}$; c. $\frac{1}{7}$; d. $\frac{2}{7}$
13. a. $\frac{1}{3}$; b. $\frac{2}{3}$; c. $\frac{2}{15}$; d. $\frac{1}{5}$
14. a. $\frac{1}{26}$; b. $\frac{5}{13}$; c. $\frac{15}{26}$; d. $\frac{1}{26}$
15. a. 0.403; b. 0.597; c. 0.00174; d. 0.291
16. a. $\frac{1}{22}$; b. $\frac{125}{1728}$; c. $\frac{5}{144}$
17. a. $\frac{10}{31}$; b. $\frac{15}{31}$; c. $\frac{6}{31}$
18. a. $\frac{1}{60}$; b. $\frac{3}{5}$; c. $\frac{1}{5}$; d. $\frac{2}{5}$; e. $\frac{1}{20}$; f. $\frac{3}{5}$; g. $\frac{3}{10}$; h. $\frac{9}{10}$; i. $\frac{1}{10}$; j. $\frac{3}{5}$
19. a. $\frac{3}{8}$; b. $\frac{1}{2}$; c. $\frac{21}{32}$; d. $\frac{3}{32}$; e. $\frac{17}{64}$
20. a. $\frac{281}{462}$; b. 8
21. a. $\frac{1}{27417}$; b. $\frac{28}{703}$
22. a. $\frac{833}{128}$; b. $\frac{198}{4165}$; c. $\frac{88}{4165}$; d. $\frac{1}{4165}$; e. $\frac{6}{4165}$; f. $\frac{1}{32487}$; g. $\frac{16660}{16660}$; h. $\frac{1}{649740}$
23. a. $\frac{1}{125}$; b. $\frac{4}{125}$; c. $\frac{16}{125}$; d. $\frac{108}{125}$
24. a. $\frac{48}{125}$; b. $\frac{n^2(n-1)^2(n-2)!}{2n^n}$